



Mobile Developer Community Meetup

March 11, 2026



Agenda

- Land Acknowledgement
- DevX Team Updates
- Apple and Google Program Updates
- Presentation by Branko Bajic: Certificate Pinning in Mobile Apps



Land Acknowledgement

- I would like to acknowledge the land on which I am privileged to work, live and recreate on is the traditional territory of the Malahat people.



DevX Team Updates

- Charles has joined the DevX team as PO (Product Owner).
- He is taking ownership of the Apple Enterprise and Google Play Store accounts from Jen, who is moving to Director of Private & Public Cloud Services
- Shiho will remain owner of the Apple Developer account



DevX Team Updates – Community Meetups

- The DevX, Public Cloud, and Private Cloud teams have combined their various meetings into one “Cloud Services Community Meetup”
- It is a meetup every 6 weeks where the 3 teams share one space to connect, collaborate, and exchange updates.
- The first one was held yesterday (March 10th)



DevX Team Updates – Community Meetups

- Question for the community: **Do you want to keep the mobile community meetup as its own meetup or combine it with the “Cloud Services Community Meetup”?**
- Please complete the “Mobile Developer Community Meetup – Survey” and share your thoughts!
- Survey open now through April 1st



Questions

- Any questions?



Apple Program Updates

- Apple Enterprise Program was renewed on February 3rd, 2026
 - Audit of test devices associated with the account will be done May or June
- Apple Developer Program renews on April 14, 2026.
 - DevX team is currently in the process of renewing it



Google Updates - Android Developer Verification

- Starting in September 2026, Android apps must be registered to a developer with a verified identity in order to be installed by users on certified Android devices.
- It requires:
 - Identity verification
 - Package name registration



Google Updates - Android Developer Verification

- Identity verification - DONE
- Package name registration
 - Starting in March, Google will automatically register Play apps
 - If it can't auto register your app, you can manually verify ownership of the app's private signing key
- September enforced in Brazil, Singapore, Indonesia, and Thailand
- 2027 and beyond – rolled out globally



Questions

- Any questions?



Presentation - Certificate Pinning in Mobile Apps

Please welcome Branko Bajic

Certificate Pinning in Mobile Apps

Branko Bajic

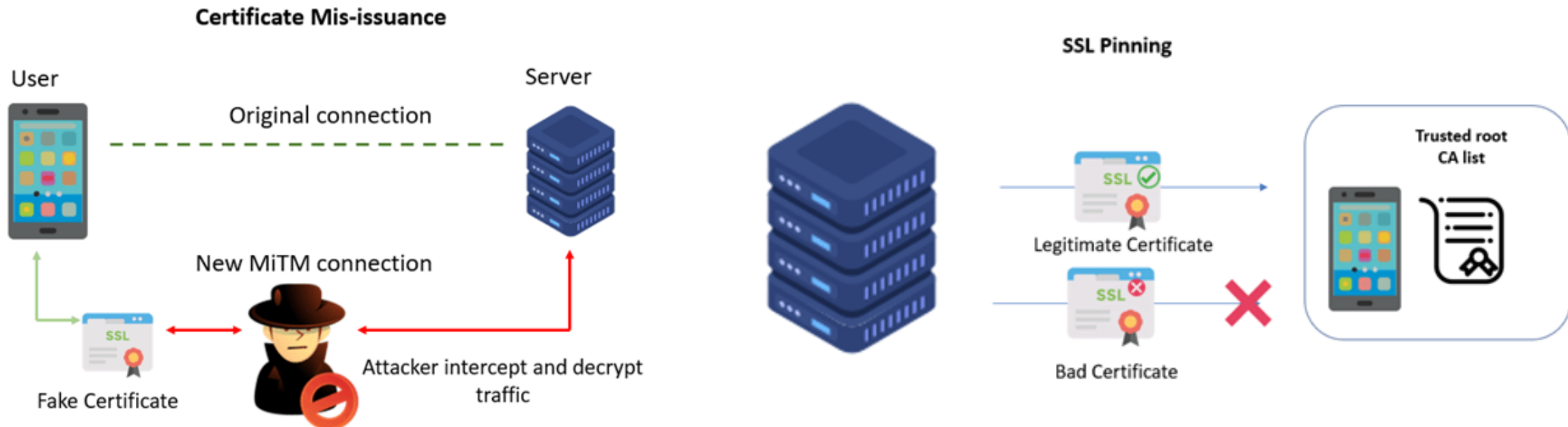
Context

- All STRA risks addressed aside from MITM attack potential
- I investigated the possibility of implementing certificate pinning in our mobile app to address this
- The intention of this presentation is to summarize what I've found and hopefully gather feedback on my proposed plan

What is Certificate Pinning?

<https://www.indusface.com/learning/what-is-ssl-pinning-a-quick-walk-through/>

- It's possible for an attacker to spoof a certificate and intercept data (MITM attack)
- Certificate pinning prevents this by hardcoding a specific certificate or public key into the app

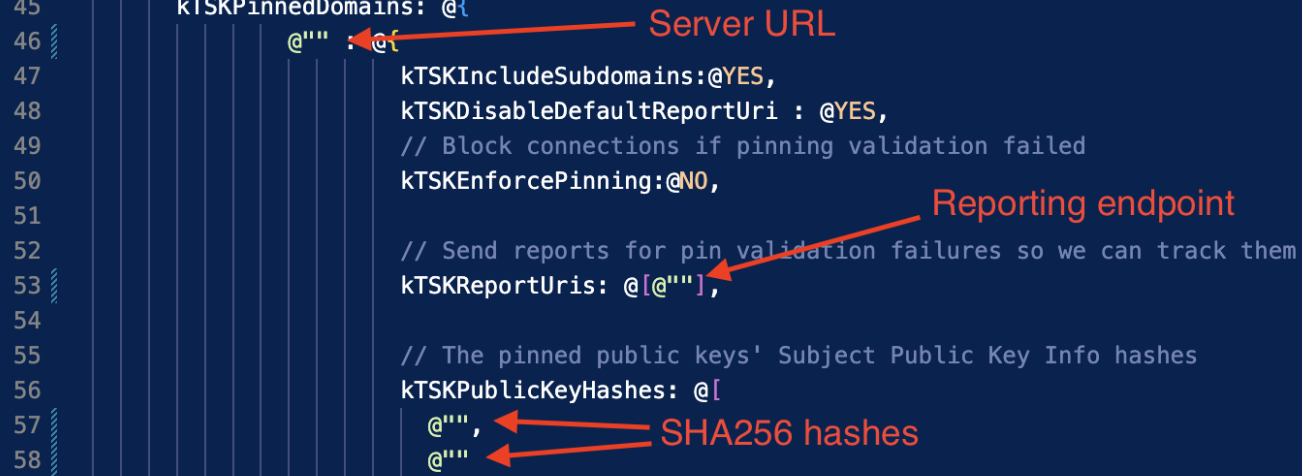


iOS Implementation

AppDelegate.mm

<https://github.com/datatheorem/TrustKit/blob/master/docs/getting-started.md>

```
36 void (^loggerBlock)(NSString *) = ^void(NSString *message) {
37     NSLog(@"TrustKit log: %@", message);
38 };
39 [TrustKit setLoggerBlock:loggerBlock];
40 NSDictionary *trustKitConfig =
41     @{
42         kTSKSwizzleNetworkDelegates : @YES,
43
44         // The list of domains we want to pin and their configuration
45         kTSKPinnedDomains: @{
46             @"": {
47                 kTSKIncludeSubdomains:@YES,
48                 kTSKDisableDefaultReportUri : @YES,
49                 // Block connections if pinning validation failed
50                 kTSKEnforcePinning:@NO,
51
52                 // Send reports for pin validation failures so we can track them
53                 kTSKReportUri: @"",
54
55                 // The pinned public keys' Subject Public Key Info hashes
56                 kTSKPublicKeyHashes: @[
57                     "",
58                     ""
59                 ]
60             }
61         };
62     [TrustKit sharedInstanceWithConfiguration:trustKitConfig];
63     [TrustKit sharedInstance].pinningValidatorCallback = ^(
64         TSKPinningValidatorResult *result, NSString *notedHostname,
65         TKSDomainPinningPolicy *policy) {
```



Android implementation

<https://square.github.io/okhttp/3.x/okhttp/okhttp3/CertificatePinner.html>

SSLPinningFactory.kt

```
1 import com.facebook.react.modules.network.OkHttpClientFactory
2 import com.facebook.react.modules.network.OkHttpClientProvider
3 import okhttp3.CertificatePinner
4 import okhttp3.OkHttpClient
5
6 class SSLPinningFactory : OkHttpClientFactory {
7     override fun createNewNetworkModuleClient(): OkHttpClient {
8         val hostname = "" ← Server URL
9
10        val certificatePinner = CertificatePinner.Builder()
11        |   .add(hostname, "") ← SHA256 hash
12        |   .build()
13
14        val clientBuilder = OkHttpClientProvider.createClientBuilder()
15
16        return clientBuilder
17        |   .certificatePinner(certificatePinner)
18        |   .build()
19    }
20 }
```

MainActivity.kt

```
11 import android.os.Bundle
12 import com.facebook.react.modules.network.OkHttpClientProvider
13 import SSLPinningFactory
14
15 class MainActivity : ReactActivity() {
16     override fun onCreate(savedInstanceState: Bundle?) {
17         super.onCreate(null);
18         OkHttpClientProvider.setOkHttpClientFactory(SSLPinningFactory());
19     }
20 }
```

The Problem

- Certificates need to be renewed
- Since the certificate is hardcoded into the app, an app update is required in order to change it
- Can't get certificates in advance
- OCIO is reducing the maximum cert lifetime (47 days by mid 2029)
- This would mean a forced app update would be required every 47 days => not ideal

A Potential Solution

Certificate Pinning vs Public Key Pinning

How does direct certificate pinning VS public key pinning make a difference here in any way ?

When using certificate pinning a new mobile app needs to be released and users forced to update each time the server certificate is rotated, while with public key pinning no need for this unless the private/public key pair used to sign the certificate changes. For example, if your server uses LetsEncrypt for the certificates you don't need to release a new mobile app version each time they are renewed.

- We could instead pin the public key
- Current replacement frequency required for public trust is every 398 days
- This would mean a forced app update to maintain pinning would only be required every 398 days
- Currently awaiting response from OCIO on whether this would be feasible



Next meetup

- **Please fill out the survey!**
- June 10th (if the community wishes to keep this meetup)
- Please contact the Developer Experience team if you would like to present!
- Developer.Experience@gov.bc.ca